CONCEPT DOCUMENT

Landscape Health Monitoring at the Humber Arboretum

Two Tools, One System

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1. Concept Overview

Our concept for the Humber Arboretum landscape health monitoring platform is built around one simple idea: the right tool for the right person, in the right place.

We noticed early on that the people who need this platform fall into two very different groups. Sustainability staff sit at a desk and need to dig into data, run reports, and track progress over time. Visitors and field workers are outside, on their phones, and just need quick access to information or a simple way to log what they see. One tool cannot serve both groups well, so we are not trying to make one.

Staff Dashboard
Web - Desktop / tablet
For sustainability staff who need to monitor data, track progress, and generate reports.

⇄ *synced*

Field App
Mobile - Works offline
For visitors and on-site staff who need quick info or want to log what they see.

Both tools share the same data. When a visitor logs an observation in the field app, it appears in the staff dashboard straight away. No manual transfer, no duplicate work.

Staff Dashboard
Web-based - For sustainability staff - Desktop / tablet

- Live health dashboards showing ecological KPIs for the arboretum
- Interactive map for seeing green vs paved areas at a glance
- Automatically generate ESG-compliant sustainability reports
- Add or import data with built-in validation
- Track how green space vs. hardscape changes over time

Field App
Mobile-first - For visitors & on-site staff - Works offline

- Look up plants and animals on the spot
- Scan QR codes placed on trees or zone markers to learn more
- Submit what you observe in just a few taps
- Browse the arboretum map and explore zones
- See sustainability updates shared by the Humber team

2. Target Audience

This platform is designed for two groups of people. They have different needs, so they each get their own tool but both tools are connected.

Primary: Sustainability Staff	Secondary: Public Visitors
Humber Office of Sustainability	*Visitors, hobbyists & tourists*
They work on campus and are responsible for tracking the arboretum's ecological health and producing sustainability reports.	They visit the arboretum to explore, learn, or observe nature. Some want to contribute their sightings to something bigger.

3. Problem Statement

This concept is a direct response to two problems, one for each audience.

Problem 1: Sustainability Staff

The Humber Office of Sustainability has no central system to track, record, and analyse landscape health data across the arboretum and campus. Without this, they cannot measure progress toward Net-Zero 2029, spot gaps in their sustainability work, or produce reliable reports for institutional leadership and funders.

Problem 2: Public Visitors

Visitors to the arboretum have no easy way to learn about the plants, wildlife, and sustainability efforts around them. There is also no way for them to contribute what they observe. This means a big opportunity for community engagement and citizen science is being missed.

Our chosen direction

How might we design a data reporting system that automatically turns landscape health data into ESG-compliant sustainability reports, while also giving visitors a meaningful way to engage with and contribute to the arboretum?

5. User Needs

Based on our research, here is what each audience needs the platform to do for them.

Sustainability Staff: what they need
See everything in one place All landscape data from the arboretum and campus in one central system, no more spreadsheets and disconnected files.
Know if they are on track A clear view of progress toward Net-Zero 2029, with real numbers they can trust and show to leadership.
Generate reports without extra work Sustainability reports that build themselves from the data already in the system, aligned with ESG and STAR Platinum standards.
Track change over time Historical data showing how green space vs. paved areas have changed, so they can demonstrate measurable impact.
Trust the data Validation checks that catch errors before they make it into reports.

Public Visitors: what they need
Understand what they are looking at Easy access to information about the plants, trees, and wildlife around them, without needing to be a scientist.
Feel like their contribution matters A simple way to log what they observe, knowing it actually feeds into something real.
Explore at their own pace A map they can browse freely, with information that appears when they want it, not all at once.
Stay connected on the go An app that works on their phone, even in areas with poor signal.

6. Key Features

The table below shows the main features of each tool and which user need they address.

Feature	Tool	User need it addresses
Live ecological health dashboard	**Staff Dashboard**	*See everything in one place*
Interactive hardscape vs. softscape map	**Staff Dashboard**	*Track change over time*
Automated ESG sustainability reports	**Staff Dashboard**	*Generate reports without extra work*
Data entry & import with validation	**Staff Dashboard**	*Trust the data*
Progress tracking toward Net-Zero 2029	**Staff Dashboard**	*Know if they are on track*
Species & plant lookup	**Field App**	*Understand what they are looking at*
QR code scanning on trees & zones	**Field App**	*Understand what they are looking at*
Quick observation submission	**Field App**	*Feel like their contribution matters*
Arboretum map & zone explorer	**Field App**	*Explore at their own pace*
Offline mode	**Field App**	*Stay connected on the go*
Profile Badges	**Field App**	*Stay motivated to collect data*

7. User Flow

Here is how each user type moves through the system from start to finish.

Sustainability Staff - Staff Dashboard

Log in to the web dashboard on their computer or tablet.

Check the ecological health overview, KPIs, alerts, and recent observations.

Open the interactive map to explore specific zones and landscape elements.

Drill into a zone to view detailed data, historical trends, or flag an issue.

Add new data manually or import a dataset, with automatic validation.

Generate a sustainability report, the system pulls the data and formats it automatically.

Download or share the report with institutional leadership or funders.

Public Visitor: Field App

1. Open the app and see a map of the arboretum.
2. Tap a zone or tree to read about the species or ecological features there.
3. Scan a QR code on a physical sign or tree marker for more detail.
4. Log an observation, take a photo, add a note, and submit in a few taps.
5. See a confirmation that their observation was received and will be reviewed.
6. Browse sustainability updates shared by the Humber team.

8. Design Principles

These principles guide every design decision we make on this project.

Accessibility first
Anyone should be able to use this whether they are a sustainability professional or a first-time visitor with no technical background. Features should be easy to find without needing training.

Data you can trust
All data from sensors, manual entry, or community submissions, must be validated before it is stored or used in reports. Accuracy is not optional when institutional reporting is involved.

Be honest and transparent
Metrics, progress, and data sources should always be clearly shown. The platform should never hide gaps, uncertainty, or missing data.

Built to grow
The system should be modular. Admins need to be able to add new landscape types, metrics, and data sources as the arboretum's needs change over time.

Make contribution feel worthwhile
When visitors submit an observation, they should feel like it actually counts. The pathway from 'I saw something' to 'it's in the system' should be quick and rewarding.

Sustainable by design
The platform itself should be lightweight, low-cost, and easy to maintain, reflecting the same values it is designed to support.

9. Out of Scope

To keep this project realistic and focused, the following are deliberately not part of this concept. These could be explored in future phases.

Not included in this concept	Why
Automated IoT sensor integration	*Hardware availability is unknown. Manual data entry covers the baseline.*
Native iOS or Android apps	*A mobile web app is sufficient for the field app in this phase.*
Multi-language support	*Out of scope for this semester. English only for now.*
Real-time alerts or push notifications	*Not required for the core reporting use case at this stage.*
Third-party API integrations (e.g. iNaturalist)	*Possible future enhancement, but adds complexity beyond this project's scope.*

10. Why We Chose This Concept

We looked in several different directions before landing on this one. Here is why the two-tool split made the most sense for this project.

It matches how people actually use the platform

Sustainability staff and public visitors have completely different needs, contexts, and devices. Trying to squeeze both into one tool would mean making trade-offs that hurt both groups. Splitting the tools means each one can be built properly for the people using it.

It keeps things simple for visitors

Public visitors should not have to navigate a complex data platform just to look up a tree or log a sighting. A lightweight mobile app with a clear purpose is much more likely to get used and used well.

It gives staff the depth they need

The Office of Sustainability needs real tools: historical trends, data validation, ESG reporting, hardscape vs. softscape tracking. A mobile-first app cannot deliver this. A dedicated web dashboard can.

It connects both groups through shared data

Even though the tools are separate, they are part of one system. Community observations from the field app feed directly into the staff dashboard. This means citizen science contributions actually count toward institutional reporting, which is rare and valuable.

It is realistic to build and maintain

Two focused tools are easier to scope, build, and hand over than one overloaded platform. This matters especially for the Humber Office of Sustainability, who need something they can maintain without dedicated IT support.

In short

This concept gives every type of user exactly what they need, nothing more, nothing less. Staff get powerful tools for monitoring and reporting. Visitors get a simple, enjoyable way to engage with the arboretum. And the data flows between them automatically.